

Week – 12 Project

MongoDB Exercise 2

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1. Create Operation

- Create a new database named "library" and switch to it.
- Create a collection named "books" with fields: title, author, and published_year.
- Insert a document into the "books" collection with the details of your favorite book.

```
C:\Users\dhanw>mongosh
Current Mongosh Log ID: 69f7867456d0d15a90abc113
Connecting to:   mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:  8.2.7
Using Mongosh:  2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
2026-05-03T20:48:21.049+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use library
switched to db library
library> db.createCollection("books")
{ ok: 1 }
library> db.books.insertOne({
  |   title: "Harry Potter and the Philosopher's Stone",
  |   author: "J.K. Rowling",
  |   published_year: 1997
  | })
{
  acknowledged: true,
  insertedId: ObjectId('69f7868a56d0d15a90abc114')
}
library> db.books.insertMany([
  |   { title: "1984", author: "George Orwell", published_year: 1949 },
  |   { title: "The Catcher in the Rye", author: "J.D. Salinger", published_year: 1951 },
  |   { title: "MongoDB Basics", author: "John Doe", published_year: 2022 },
  |   { title: "Learning NoSQL", author: "Jane Smith", published_year: 2021 }
  | ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f7869556d0d15a90abc115'),
    '1': ObjectId('69f7869556d0d15a90abc116'),
    '2': ObjectId('69f7869556d0d15a90abc117'),
    '3': ObjectId('69f7869556d0d15a90abc118')
  }
}
```

2. Read Operation

- Retrieve all documents from the "books" collection.
- Find and display only the documents where the author is "J.K. Rowling".
- Fetch and display the document with the earliest published year.

```

}
library> db.books.find().pretty()
[
  {
    _id: ObjectId('69f7868a56d0d15a90abc114'),
    title: "Harry Potter and the Philosopher's Stone",
    author: 'J.K. Rowling',
    published_year: 1997
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc115'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 1951
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc118'),
    title: 'Learning NoSQL',
    author: 'Jane Smith',
    published_year: 2021
  }
]

```

```

library> db.books.find({ author: "J.K. Rowling" })
[
  {
    _id: ObjectId('69f7868a56d0d15a90abc114'),
    title: "Harry Potter and the Philosopher's Stone",
    author: 'J.K. Rowling',
    published_year: 1997
  }
]
library> db.books.find().sort({ published_year: 1 }).limit(1)
[
  {
    _id: ObjectId('69f7869556d0d15a90abc115'),
    title: '1984',
    author: 'George Orwell',
    published_year: 1949
  }
]
library> |

```

3. Update Operation

- Update the published year of the book with the title "The Catcher in the Rye" to the current year.
- Add a new field "genre" with the value "Mystery" to all documents in the "books" collection.

```
library> db.books.updateOne(  
|   { title: "The Catcher in the Rye" },  
|   { $set: { published_year: 2026 } }  
| )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
library> db.books.updateMany(  
|   {},  
|   { $set: { genre: "Mystery" } }  
| )  
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 5,  
  modifiedCount: 5,  
  upsertedCount: 0  
}  
library> |
```

4. Delete Operation

- Remove the document with the title "1984" from the "books" collection.
- Delete all documents from the "books" collection where the published year is before 2000.

```
library> db.books.deleteOne({ title: "1984" })  
{ acknowledged: true, deletedCount: 1 }  
library> db.books.deleteMany({ published_year: { $lt: 2000 } })  
{ acknowledged: true, deletedCount: 1 }  
library> |
```

5. Advanced Query

- Find and display the top 3 recently published books from the "books" collection.
- Retrieve documents from the "books" collection where the title contains the word "MongoDB" or "NoSQL".

```
library> db.books.find().sort({ published_year: -1 }).limit(3)
[
  {
    _id: ObjectId('69f7869556d0d15a90abc116'),
    title: 'The Catcher in the Rye',
    author: 'J.D. Salinger',
    published_year: 2026,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc118'),
    title: 'Learning NoSQL',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library> db.books.find({
  $or: [
    { title: { $regex: "MongoDB", $options: "i" } },
    { title: { $regex: "NoSQL", $options: "i" } }
  ]
})
[
  {
    _id: ObjectId('69f7869556d0d15a90abc117'),
    title: 'MongoDB Basics',
    author: 'John Doe',
    published_year: 2022,
    genre: 'Mystery'
  },
  {
    _id: ObjectId('69f7869556d0d15a90abc118'),
    title: 'Learning NoSQL',
    author: 'Jane Smith',
    published_year: 2021,
    genre: 'Mystery'
  }
]
library> |
```